Pure Neumann Boundary Conditions

Solvability, Gauge Fixing, and Reaction Resonance
in the Jacobi-Simplex HPS Solver

Mathematical and implementation design note - Sachin Natesh

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Abstract

We develop a single pure-Neumann treatment for the existing divergence-form Jacobi-simplex hierarchical Poincaré–Steklov (HPS) discretization of $-\nabla\cdot(A\nabla u)+b\cdot\nabla u+cu=f$. The distinction is spectral, not a sequence of separate diffusion, advection, and reaction implementations. For $c=0$, constants are right-null functions; solvability requires compatibility with the *adjoint* nullspace and a user-configurable gauge selects the solution. For $c\neq 0$, the constant nullspace generally disappears, so an ordinary pure-flux boundary-value problem may be invertible. Positive reaction gives a particularly favorable case. Negative reaction can destroy energy coercivity without destroying invertibility; resonance is governed by the Neumann spectrum, and general Fredholm alternatives describe exceptional singular cases. We derive the continuous weak and adjoint formulations, give a root-only HPS implementation plan, and distinguish the next-release scope from more general resonant regularization.

1 The problem and conventions

Let $\Omega\subset\mathbb{R}^D$ be a bounded, connected Lipschitz domain and set $\Gamma=\partial\Omega$. Assume $A(x)$ is bounded and uniformly elliptic; unless otherwise specified, take $A=A^T$ and

$$\xi^T A(x)\xi\geq a_0|\xi|^2,\qquad a_0>0. \tag{1}$$

The coefficients $b\in L^\infty(\Omega)^D$ and $c\in L^\infty(\Omega)$ are real. Our strong problem is

$$\boxed{\begin{aligned} Lu:=-\nabla\cdot(A\nabla u)+b\cdot\nabla u+cu&=f \quad \text{in } \Omega,\\ q:=n\cdot A\nabla u&=g \quad \text{on } \Gamma.\end{aligned}} \tag{2}$$

Here n is the *outward* unit normal. Note that the physical diffusive transport flux is often $J=-A\nabla u$, so $J\cdot n=-g$; that sign distinction matters for physical conservation statements.

The weak formulation, including the inhomogeneous prescribed flux, is

$$a_c(u,v):=\int_\Omega A\nabla u\cdot\nabla v\,\mathrm{d}x+\int_\Omega (b\cdot\nabla u)v\,\mathrm{d}x+\int_\Omega cuv\,\mathrm{d}x=F(v),\quad v\in H^1(\Omega), \tag{3}$$

with

$$F(v)=\int_\Omega fv\,\mathrm{d}x+\int_\Gamma gv\,\mathrm{d}S. \tag{4}$$

Boundary integrals are understood via traces or normal-flux duality for the usual Sobolev data classes. We keep the entire A,b,c operator throughout.

2 The nullspace and the exact compatibility condition

2.1 No reaction: $c = 0$

Since $L1 = c$, for $c = 0$ the constant function satisfies the homogeneous PDE and homogeneous co-normal condition. On a connected domain, under the standard uniform-ellipticity and regularity hypotheses, the homogeneous Neumann problem has a one-dimensional right nullspace spanned by 1 (including with bounded first-order drift). Consequently, if a solution exists then $u + C$ is also a solution.

For $b = 0$, take $v = 1$ in (3). The diffusion term vanishes and we obtain the *physical conservation constraint*

$$\boxed{\int_\Omega f \,\mathrm{d}x + \int_\Gamma g \,\mathrm{d}S = 0.} \tag{5}$$

This is necessary and, for the connected uniformly elliptic diffusion problem, sufficient for solvability modulo constants. It constrains only the total flux; it does not determine the spatial function g from f.

For nonzero drift, testing (3) with 1 instead gives $\int_\Omega b \cdot \nabla u = F(1)$, so (5) is no longer the correct data-only compatibility condition in general.

2.2 Adjoint compatibility for $b \neq 0$

The formal interior adjoint and its homogeneous boundary condition are

$$\begin{aligned} L^*\phi = -\nabla \cdot (A^T \nabla \phi) - \nabla \cdot (b\phi) + c\phi = 0 \quad &\text{in } \Omega, \\ n \cdot A^T \nabla \phi + (b \cdot n)\phi = 0 \quad &\text{on } \Gamma. \end{aligned} \tag{6}$$

Integration by parts, using the primal flux $n \cdot A\nabla u = g$, yields

$$\int_\Omega \phi L u \,\mathrm{d}x = \int_\Omega u L^* \phi \,\mathrm{d}x + \int_\Gamma u\big[n \cdot A^T \nabla \phi + (b \cdot n)\phi\big] \,\mathrm{d}S - \int_\Gamma \phi g \,\mathrm{d}S. \tag{7}$$

Thus every homogeneous adjoint null function imposes

$$\boxed{F(\phi) = \int_\Omega \phi f \,\mathrm{d}x + \int_\Gamma \phi g \,\mathrm{d}S = 0, \qquad \phi \in \ker L^*.} \tag{8}$$

For $b = c = 0$, $\phi = 1$ is the adjoint null function and this reduces to (5). For general $b \neq 0$, ϕ is usually nonconstant. *The implementation need not solve a second continuous adjoint PDE:* the discrete HPS root operator already has a left nullspace that captures its algebraic compatibility condition.

2.3 Gauge: a uniqueness condition, not a compatibility correction

For compatible data and $c = 0$, fix a linear functional $\mathcal{G}$ satisfying $\mathcal{G}(1) \neq 0$ and prescribe

$$\boxed{\mathcal{G}(u) = \eta, \qquad \eta \in \mathbb{R}.} \tag{9}$$

A convenient root-only choice is the physical boundary mean,

$$\mathcal{G}_\partial(u) = \frac{1}{|\Gamma|} \int_\Gamma u \,\mathrm{d}S. \tag{10}$$

It differs from a zero-*volume*-mean gauge but fixes the same additive constant. Allow $\eta = 0$ by default and expose η to the caller. For example, if u_* is a manufactured solution, the boundary-mean-gauged reference is $u_* + \eta - \mathcal{G}_\partial(u_*)$. The solver should *reject* incompatible input, rather than silently projecting or modifying f or g.

3 Why prescribed Neumann data can be solvable when $c \neq 0$

The phrase *pure Neumann* specifies the boundary data, not the presence of a nullspace. Indeed $L1 = c$, so a nonzero reaction term generally removes constants from the kernel. The boundary-value operator may be invertible without any special gauge or compatibility test. Invertibility is a property of the *entire operator* (coefficients, domain, and boundary condition), not a consequence of the word "Neumann."

3.1 Positive reaction: the Lax–Milgram case

Suppose first $b = 0$, $A = A^T$ satisfies (1), and

$$c(x) \geq c_0 > 0 \quad \text{a.e. in } \Omega. \tag{11}$$

Then

$$a_c(u,u) \geq a_0 \|\nabla u\|_{L^2}^2 + c_0 \|u\|_{L^2}^2 \geq C \|u\|_{H^1}^2, \qquad C = \min(a_0, c_0) > 0. \tag{12}$$

The form is bounded and coercive on all of $H^1(\Omega)$; Lax–Milgram gives a unique weak solution for each continuous functional F, with *no* compatibility condition and *no* gauge. If $c \geq 0$ is not identically zero, diffusion plus a reaction region of positive measure also removes the constant kernel; a generalized Poincaré inequality yields coercivity under standard hypotheses, even if $\operatorname{ess\,inf} c = 0$.

With advection $b \neq 0$, the bilinear form is nonsymmetric. A direct sufficient Lax–Milgram condition follows from Young's inequality:

$$\left| \int_\Omega (b \cdot \nabla u) u \, \mathrm{d}x \right| \leq \|b\|_{L^\infty} \|\nabla u\|_{L^2} \|u\|_{L^2} \tag{13}$$

$$\leq \frac{a_0}{2} \|\nabla u\|_{L^2}^2 + \frac{\|b\|_{L^\infty}^2}{2a_0} \|u\|_{L^2}^2, \tag{14}$$

so coercivity is immediate if

$$c_0 > \frac{\|b\|_{L^\infty}^2}{2a_0}. \tag{15}$$

This is sufficient, not necessary. For regular, uniformly elliptic scalar operators with $c \geq c_0 > 0$, the maximum principle and the Neumann boundary point principle provide uniqueness even when (15) fails; Fredholm existence then gives well-posedness. Thus positive reaction is not intrinsically troublesome for the general A, b, c solver.

4 Negative reaction and the Neumann spectral threshold

4.1 The correct indexing of Neumann eigenvalues

Here the natural spectral analysis begins with $b = 0$ and symmetric A. Let $L_0 = -\nabla \cdot (A\nabla)$ with homogeneous co-normal Neumann data. The operator is self-adjoint and nonnegative on $L^2(\Omega)$ with compact resolvent. Its eigenvalues are

$$\boxed{0 = \lambda_0^N < \lambda_1^N \leq \lambda_2^N \leq \cdots, \qquad \lambda_k^N \to \infty.} \tag{16}$$

The *smallest* Neumann eigenvalue is zero. The threshold commonly intended in a negative-reaction comparison is therefore the *first positive* eigenvalue λ_1^N, not λ_0^N. The associated weighted Poincaré–Rayleigh characterization is

$$\lambda_1^N = \inf_{\substack{v \in H^1(\Omega),\, v \neq 0 \\ \int_\Omega v \, \mathrm{d}x = 0}} \frac{\int_\Omega \nabla v \cdot A \nabla v \, \mathrm{d}x}{\int_\Omega v^2 \, \mathrm{d}x} > 0. \tag{17}$$

4.2 Constant negative reaction $c = -\mu$

Put $\mu > 0$ and set $L_\mu = L_0 - \mu I$. Its spectrum is

$$\operatorname{spec}(L_\mu) = \{\lambda_k^N - \mu : k \geq 0\}. \tag{18}$$

Hence

$$\boxed{L_\mu \text{ is invertible precisely when } \mu \notin \{\lambda_k^N : k \geq 0\}.} \tag{19}$$

In particular, every $0 < \mu < \lambda_1^N$ gives a unique solution for any admissible f, g, even though the full energy $a_{-\mu}(1,1) = -\mu|\Omega|$ is negative and Lax–Milgram *coercivity on all of H^1 fails*.

To see the useful remaining coercivity, decompose $u = m + v$ with $m = |\Omega|^{-1}\int_\Omega u$ and $\int_\Omega v = 0$. Equation (3) tested with 1 gives

$$-\mu|\Omega|m = F(1), \qquad m = -\frac{F(1)}{\mu|\Omega|}. \tag{20}$$

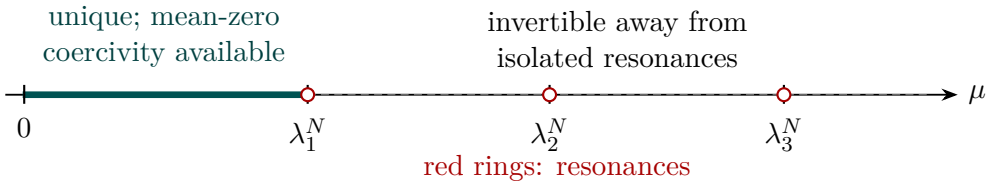
The mean-zero component solves

$$\int_\Omega A\nabla v \cdot \nabla\psi \,\mathrm{d}x - \mu\int_\Omega v\psi \,\mathrm{d}x = F(\psi) \quad \forall \psi \in H^1(\Omega),\ \int_\Omega \psi \,\mathrm{d}x = 0, \tag{21}$$

and Rayleigh–Poincaré implies

$$\int_\Omega \nabla v \cdot A\nabla v \,\mathrm{d}x - \mu\int_\Omega v^2 \,\mathrm{d}x \geq \left(1 - \frac{\mu}{\lambda_1^N}\right)\int_\Omega \nabla v \cdot A\nabla v \,\mathrm{d}x. \tag{22}$$

Therefore Lax–Milgram applies on the mean-zero subspace when $\mu < \lambda_1^N$; the constant coefficient m is determined by (20), not chosen by a gauge.



Above the threshold. For $\mu > \lambda_1^N$, the operator is *not automatically singular*. It is invertible between successive Neumann eigenvalues, but the energy has additional negative directions. At $\mu = \lambda_k^N$, a nonconstant homogeneous nullspace appears; solvability requires orthogonality of the forcing functional F to the corresponding eigenfunctions. A constant-only gauge does not handle these resonances. Near them, the inverse norm can become large, approximately like the reciprocal distance to the spectrum.

One-dimensional example. On $(0,1)$ with $A = 1$, $\lambda_k^N = k^2\pi^2$ and the normalized eigenfunctions are constants and $\cos(k\pi x)$, $k \geq 1$. For $-u'' - \mu u = f$ with $u'(0) = u'(1) = 0$ (in the outward-flux convention), every $0 < \mu < \pi^2$ is uniquely solvable. At $\mu = \pi^2$, the forcing must satisfy $\int_0^1 f(x)\cos(\pi x)\,\mathrm{d}x = 0$; when compatible, a separate condition must fix the coefficient of $\cos(\pi x)$.

4.3 Variable negative reaction

For $c(x) = -\mu\rho(x)$ with $\rho \geq \rho_0 > 0$, the appropriate thresholds are generalized eigenvalues of

$$\int_\Omega A\nabla\psi \cdot \nabla v \, \mathrm{d}x = \nu \int_\Omega \rho\psi v \, \mathrm{d}x, \qquad v \in H^1(\Omega). \tag{23}$$

The first positive generalized eigenvalue is obtained by restricting to ρ-weighted mean-zero functions and minimizing the corresponding Rayleigh quotient. As above, $0 < \mu < \nu_1$ gives an invertible operator, while singularity occurs at generalized eigenvalues. For a completely general sign-changing $c(x)$, a pointwise comparison of $c(x)$ with a single number λ_1^N is *not* a complete solvability criterion: one must analyze the full potential-perturbed operator.

4.4 What changes when $b \neq 0$?

The diffusion–advection operator $L_0 = -\nabla \cdot (A\nabla) + b \cdot \nabla$ with Neumann flux still satisfies $L_0 1 = 0$; however it is generally *nonsymmetric*. Its spectrum and adjoint eigenspaces need not admit a real Rayleigh-quotient ordering, and spectral values may be complex. A constant reaction shift $-\mu I$ still moves the spectrum, but the self-adjoint threshold λ_1^N is not available without additional symmetrizing structure. At singular values one must use the left and right kernels of the actual operator.

5 Fredholm alternative and regularization at resonance

For bounded Ω, uniformly elliptic A, and bounded lower-order coefficients, one can choose a sufficiently large $\gamma > 0$ such that $a_c(u, v) + \gamma(u, v)_{L^2}$ is coercive (by Young's inequality). The shifted solution map is bounded, and the compact embedding $H^1(\Omega) \hookrightarrow L^2(\Omega)$ yields a compact-resolvent/Fredholm formulation for the corresponding homogeneous-boundary operator. The Fredholm alternative states that either the operator is invertible, or its kernel is finite dimensional and the data must annihilate every adjoint null function. More explicitly, if

$$\ker L^* = \operatorname{span}\{\phi_1, \ldots, \phi_r\}, \tag{24}$$

with adjoint boundary conditions (6), then solvability requires

$$\boxed{F(\phi_j) = 0, \qquad j = 1, \ldots, r.} \tag{25}$$

When it holds, solutions differ by elements of $\ker L$, and r independent gauge functionals may be needed. This applies to negative and sign-changing reaction terms and, with the appropriate adjoint, to drift.

No regularization when already invertible. If a negative-reaction operator is away from resonance, solve the original problem directly. Lack of coercivity does not imply a need for Tikhonov regularization or limiting absorption.

Complex resolvent shift / limiting absorption. For a self-adjoint real operator L having a zero eigenvalue, one may consider $(L + i\varepsilon I)u_\varepsilon = F$ with $\varepsilon \downarrow 0$. For forcing orthogonal to $\ker L$, u_ε converges to the solution in $(\ker L)^\perp$; otherwise the component along the nullspace grows like $1/\varepsilon$. On a bounded domain with discrete spectrum this is most accurately viewed as a *resolvent regularization*, not as a necessary physical radiation condition. The non-self-adjoint case requires additional care (including possible nonnormality and generalized eigenspaces).

Tikhonov regularization. At the discrete level one can minimize

$$\|R\boldsymbol{\lambda} - \mathbf{h}\|_2^2 + \varepsilon^2 \|C(\boldsymbol{\lambda} - \boldsymbol{\lambda}_{\mathrm{ref}})\|_2^2, \qquad \varepsilon > 0. \tag{26}$$

As $\varepsilon \to 0$, compatible data give a selected solution according to the regularization metric and reference vector; incompatible data instead give a least-squares compromise, *not* a solution to the original PDE. We should therefore retain an explicit compatibility test before any such regularized solve. The gauge selected by Tikhonov is not automatically the user's desired physical mean gauge.

6 Root-only integration with the existing HPS solver

6.1 What is already available

The Dense divergence-form HPS solver represents an exterior trace vector $\boldsymbol{\lambda} \in \mathbb{R}^{N_B}$ in a common canonical face Jacobi basis and supplies an affine numerical co-normal flux map

$$\widehat{\boldsymbol{\mu}} = S_{\mathrm{root}}\boldsymbol{\lambda} + \mathbf{b}_{\mathrm{root}}. \tag{27}$$

The map S_{root} comes from the existing tau-stabilized leaf and merge algebra; $\mathbf{b}_{\mathrm{root}}$ is updated from the interior source. Pure Neumann with $\alpha = 0$, $\beta = 1$ therefore gives

$$\boxed{R\boldsymbol{\lambda} = \mathbf{h}, \qquad R = S_{\mathrm{root}}, \qquad \mathbf{h} = \mathbf{g} - \mathbf{b}_{\mathrm{root}}.} \tag{28}$$

When $\beta = 1$ is constant, the higher-degree co-normal flux correction M_β^{LH} is zero by orthogonality, so no additional high-mode term is required. The co-normal operator already incorporates variable A and interface flux continuity.

For $\alpha = 0$ but spatially varying, nowhere-vanishing β, the already-implemented corrected variable-Robin equation instead uses

$$R = M_\beta^{LL} S_{\mathrm{root}} + M_\beta^{LH} H_{\mathrm{root}}, \qquad \mathbf{h} = \mathbf{g} - M_\beta^{LL}\mathbf{b}_{\mathrm{root}} - M_\beta^{LH} z_{\mathrm{root}}. \tag{29}$$

The same nullspace methodology applies. If β vanishes, the boundary operator may change type and requires a separate well-posedness analysis.

6.2 Singular constant-nullspace case, usually $c = 0$

Let $v \in \mathbb{R}^{N_B}$ be the coefficient vector of the *constant exterior trace*. For an exact singular pure-Neumann root map, $Rv = 0$. Numerically Rv will only be small, so assess its residual against the operator scale and verify the expected rank deficiency rather than declaring all ill-conditioned operators to have a constant nullspace. Find a left-null vector ℓ (e.g. using an SVD or a suitably rank-revealing factorization):

$$Rv \simeq 0, \qquad R^T\ell \simeq 0, \qquad \ell^T\ell = 1. \tag{30}$$

The discrete compatibility condition is

$$\boxed{\ell^T\mathbf{h} \simeq 0.} \tag{31}$$

Use a scaled, conditioning-aware tolerance, report the normalized defect, and reject inconsistent data. The left vector ℓ need not coincide with the constant trace or with a coefficient-wise vector of ones, especially in nonsymmetric, weighted coordinates.

Construct a root boundary-mean row w^T using physical face integration and normalize it such that $w^T v = 1$. The caller supplies η and the gauge is $w^T \boldsymbol{\lambda} = \eta$. A convenient bordered root system is

$$\boxed{\begin{bmatrix} R & \ell \\ w^T & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \sigma \end{bmatrix} = \begin{bmatrix} \mathbf{h} \\ \eta \end{bmatrix}.} \tag{32}$$

If R has precisely a one-dimensional right and left nullspace and $w^T v \neq 0$, this system is nonsingular. For compatible data, σ is zero to numerical tolerance; σ must not be used silently to absorb an incompatible RHS. The gauge affects only the constant component and does not alter the physical gradient/flux. No changes to leaf solves, interface merges, or high co-normal observations are needed.

6.3 What happens for $c \neq 0$?

For an ordinary well-posed pure-Neumann problem with reaction, use the existing nonsingular root solve, including the current LU factorization. *Do not impose a constant gauge merely because* $\alpha = 0$. If the root matrix is singular or nearly singular at a reaction resonance, report the condition/rank diagnostic. Its null modes may be nonconstant; proper treatment would require a generalized left/right-nullspace procedure and is outside the near-term singular-Neumann implementation.

6.4 Proposed solver contract

1. Assemble the ordinary Dense divergence-form HPS tree and root flux map.
2. Accept $\alpha = 0$, $\beta = 1$ on all exterior faces as a legitimate *boundary specification*, regardless of b and c.
3. For a verified constant-nullspace case ($c = 0$ within a deliberate coefficient-identification rule), construct the right constant trace, compute/validate the root left nullspace, and check (31).
4. If compatible, apply the boundary-mean gauge $w^T \boldsymbol{\lambda} = \eta$ using (32); default $\eta = 0$.
5. If the root operator is nonsingular (notably when $c > 0$ and the problem is well posed), use the normal solve and no gauge.
6. If an unanticipated nonconstant nullspace/resonance is detected, report it rather than assigning a spurious constant gauge.

A special explicit `pure_neumann` flag could guard the singular branch; for general α, β the existing Robin code remains authoritative. The degree-n HPS skeleton, tau parameter, and Dense-only divergence-form restriction remain unchanged.

7 Validation plan and release boundary

The first implementation can exercise the *general* A, b, c operator without staging different PDE solvers. Suggested tests on the already used conforming simplex decompositions of the unit cube are summarized below.

Test	Coefficients and data	Required outcome
A	A cubic SPD, $b = c = 0$; degree-5 polynomial u_*, exact f, g	Compatibility accepted; $u_* + C$ recovered to numerical precision under prescribed boundary-mean gauge.
B	Same A; nonzero polynomial b, $c = 0$	Generalized *left-null* compatibility accepted; exact polynomial recovery under the gauge.
C	As B with $c(x) \geq c_0 > 0$	Nonsingular ordinary root solve with no gauge.
D	Perturb only g by a nonzero constant in A or B	Compatibility rejected with a diagnostic (no silent projection).
E	Repeat A or B with two gauge right-hand sides η_1, η_2	Solutions differ by the constant $\eta_2 - \eta_1$ when $w^T v = 1$; gradients and fluxes unchanged.
F (later)	$b = 0$, constant $c = -\mu$ near/at λ_k^N	Invertible away from resonance; reject or separately regularize resonant cases.

The current mixed Neumann/Dirichlet/variable-Robin manufactured-solution regressions already recover the degree-5 global polynomial to roundoff in $D = 1, 2, 3$ and require *no* nullspace machinery. The specific addition here is the *all-Neumann* singular branch, together with a nonsingular reaction case and informative treatment of exceptional resonances.

8 Conclusions

1. Pure Neumann means prescribed co-normal flux on every exterior face. It does *not* inherently mean that the operator is singular.

2. If $c = 0$, constants are right-null modes. For general drift b, test compatibility against the *left/adjoint* nullspace, then impose a caller-configurable gauge, preferably the boundary mean at the root.

3. A strictly positive reaction makes the diffusion problem coercive; with drift, maximum-principle and Fredholm arguments cover cases beyond the simple sufficient Lax–Milgram estimate.

4. For negative constant reaction $c = -\mu$ in the self-adjoint case, the first positive Neumann eigenvalue λ_1^N marks the loss of coercivity on mean-zero functions. Invertibility fails only at the discrete resonances $\mu = \lambda_k^N$; there is no blanket ill-posedness for $\mu > \lambda_1^N$.

5. Fredholm compatibility, resonant mode gauges, complex shifts, and Tikhonov methods provide a principled route to broader indefinite problems, but need not block the next release's nonresonant Neumann implementation.